

JECRC University, Jaipur
Department of Physics
B. Tech Ist Year – Ist Semester July-Dec 2025-26
Applied Physics (DPH001B)
Assignment-V

- A. Very Short Answer Questions (10)**
1. State Gauss divergence theorem for vectors? (2)
 2. Write the mathematical statement of Gauss theorem for magnetostatics. (2)
 3. Does Electromagnetic waves require medium to travel? (2)
 4. For Any given vector $\mathbf{A}$, $\nabla \times \mathbf{A} = \mathbf{0}$. Is $\mathbf{A}$ rotational? (2)
 5. Why EM waves are termed as transverse waves? (2)
- B. Answer the following briefly (21)**
1. Prove that $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{A}) = 0$ (7)
 2. Write Ampere's Law and what is Maxwell's correction to that Law. (7)
 3. If earth receives 2-cal/min-cm^2 of solar energy. What is magnitude of electric and magnetic field of solar radiation? (7)
- C. Answer the followings explicitly. (33)**
1. Using differential form of Maxwell's equations derive the EM wave equation in vacuum. (11)
 2. Write differential form and integral form of Maxwell's equations. Explain the physical significance of each equation. (11)
 3. Write Short notes on a) Poynting Theorem b) Stokes Theorem (6+5)

Solution: As Poynting vector,

$$\begin{aligned}\vec{S} &= \vec{E} \times \vec{H} \\ &= \vec{E}\vec{H} \sin 90^\circ \\ &= \vec{E}\vec{H}\end{aligned}$$

$$\text{Solar energy} = 2 \text{ cal/min-cm}^2$$

$$= \frac{2 \times 4.18 \times 10^4}{60} \text{ Jm}^{-2}\text{s}^{-1}$$

Both are energy flux per unit area per second.

Hence,

$$\vec{E}\vec{H} = \frac{2 \times 4.18 \times 10^4}{60} \approx 1400$$

But

$$\frac{\vec{E}}{\vec{H}} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi = 377$$

$\therefore$

$$\vec{E}\vec{H} \times \frac{\vec{E}}{\vec{H}} = 1400 \times 377$$

$$\vec{E} = \sqrt{1400 \times 377} = 726.5 \text{ V/m}$$

Now,

$$\vec{H} = \frac{E}{377} = 1.927 \text{ Amp/m}$$

Amplitudes of electric and magnetic field of radiation are

$$\vec{E}_0 = \vec{E}\sqrt{2} = 1024.3 \text{ V/m}$$

and

$$\vec{H}_0 = \vec{H}\sqrt{2} = 2.717 \text{ Amp/m}$$